Conjecture 00000000721: negative factorial powers do not give an uncountable p-adic family

lizaixi01, with OpenAI Codex assistance

22 October 2026

1 The stated series and its ambient field

The source claims that the set

$$\left\{ \sum_{n\geq 1} a_n p^{-n!} : a_n \in \{0, \dots, p-1\} \right\}$$

contains an algebraically independent subfamily of cardinality $2^{\aleph_0}$ in $\overline{\mathbf{Q}}_p$. We retain the negative exponent exactly as stated and use the usual extension of the p-adic absolute value, normalized by $|p|_p = p^{-1}$. At $p = 2$, every convergent series in this set is a finite sum and is rational. Its value set is therefore countable, which excludes the required cardinality regardless of algebraic independence.

The ambient field in the proof is the actual algebraic closure $K = \overline{\mathbf{Q}}_2$ of $\mathbf{Q}_2$, equipped with its extended 2-adic norm. Completeness is unnecessary: we consider series that actually converge in K. Changing the sign in the exponent would produce a different claim, to which this argument does not apply.

2 Convergence forces finite support

Let $a_n \in \{0, 1\}$. If $a_n \neq 0$, then $a_n = 1$ and

$$|a_n 2^{-n!}|_2 = 2^{n!} \geq 1. \tag{1}$$

In every normed additive group, convergence of the sequence of partial sums implies that the individual terms tend to zero: subtract the partial sum ending at $n-1$ from the one ending at n. It follows that eventually $|a_n 2^{-n!}|_2 < 1$. By (1), all sufficiently late digits are zero.

Consequently there is N such that every convergent value is

$$\sum_{n=1}^{N} a_n 2^{-n!} \in \mathbf{Q}.$$

The rational numbers are countable, and the field embedding $\mathbf{Q} \hookrightarrow K$ carries a countable set to a countable set. Thus the full set of actual series values is countable. Every subset is countable, whereas $2^{\aleph_0} > \aleph_0$ by Cantor's theorem. The specified continuum-sized subfamily cannot exist.

3 The two meanings of series convergence are covered

The code first proves the finite-support statement for Lean's `HasSum` predicate. It then explicitly treats the ordinary sequence of forward partial sums

$$s_N = \sum_{j=0}^{N-1} a_{j+1} 2^{-(j+1)!}.$$

If $s_N \to x$, then $s_{N+1} - s_N \to 0$. Applying (1) to the terms shows that the digits eventually vanish. The resulting finite sum has a `HasSum` proof, and uniqueness of the partial-sum limit identifies its value with x. Conversely, `HasSum` implies convergence of these forward partial sums. Hence the two actual value sets coincide in this example. No value is assigned to any divergent series.

The formal term sequence is indexed from zero, with term number j equal to $a_{j+1}2^{-(j+1)!}$ in the source indexing. Both this equality and the surjectivity of the digit reindexing are proved, so no zeroth factorial term has been inserted into the source series.

4 Formal objects and verification

`K` is Mathlib's `PadicAlgCl 2`, the algebraic closure with the spectral norm extending the 2-adic norm. `norm_two` uses that extension and the normalization on $\mathbf{Q}_2$. `norm_term_of_nonzero` proves the lower bound (1), and the vanishing-term argument is in `vanishing_eventually_zero`. `series_value_rational` constructs the finite rational sum explicitly and proves that its field embedding is the actual sum value.

`sequential_hasSum` and `sourceSequentialValues_eq` establish the ordinary partial-sum bridge. The final theorem `no_continuum_subfamily` directly negates the existence of a subset of the source value set whose `Cardinal.mk` equals $2^{\aleph_0}$. It therefore refutes a necessary condition of the original claim without assuming anything about the transcendence degree.

The project pins Lean 4.33.0, Mathlib and all dependency commits. A new root project build, fourteen-declaration transitive axiom audit and official fresh-kernel replay are recorded with source hashes. No admitted proof, native decision shortcut or extra axiom is used. The original bilingual source is preserved from upstream commit `b355e42906e6eaf52bc7e9d005e0ef02d8aa7400`.

OpenAI Codex assisted the mathematics, proof and report. A separate supervising Codex agent reviewed the sign, topology, indexing and both convergence interpretations. This is AI review, not human certification. The argument uses elementary necessary conditions for convergence and countability; no new mathematical result is claimed.